

Qijun Wu

+86 13560800316 | wuqijun1001@163.com | github.com/01seven

EDUCATION

Shenzhen University <i>Incoming M.S. Student in Applied Statistics</i>	Expected enrollment: Sep. 2027
Guangzhou University <i>B.S. in Statistics</i>	Sep. 2023 - Jun. 2027 (Expected)
• Academic performance: GPA 3.78/4.00; average score 90.73; ranked 7th of 72 students (top 10%); CET-6. • Relevant coursework: Probability Theory (91), Mathematical Statistics (91), Statistical Computing (96), Multivariate Statistics (93), Data Mining (92), Regression Analysis (98), Time Series Analysis (91), Data Structures & Algorithm Analysis (99), Mathematical Analysis Topics (98), and Advanced Algebra Topics (99).	

RESEARCH INTERESTS

My research interests lie at the intersection of **AI for Science (AI4S)**, **AI for Statistics** and **High-Dimensional inference**. I am particularly interested in AI-assisted scientific computing and GPU-based statistical simulations, including their validation. My statistical research focuses on **high-dimensional inference** and **interval estimation**, with attention to Type I error control, test power, coverage, and interval length through theory and simulation.

RESEARCH EXPERIENCE

Power-Enhanced Tests for the High-Dimensional Two-Sample Behrens-Fisher Problem

- Developed **WCCT** by combining Welch's t -tests with the Cauchy combination method, and proposed **EWCCCT** and **EWCCTS** as power-enhanced variants for sparse, dense, and scale-heterogeneous signals.
- Designed simulation studies and real-data analyses to compare **Type I error control** and **empirical power** under small-to-moderate sample sizes.
- Led method development, theoretical derivations, experimental design, result analysis, and manuscript writing.

Prior-Free Probabilistic Interval Estimation for Constrained Parameters

- Developed prior-free **IM** and **NIM** intervals for constrained Normal and Poisson means with nuisance parameters.
- For Normal means, **IM** had exact coverage with slightly longer intervals than Bayesian methods; for Poisson means, **IM** was **conservative** and **NIM** coverage was **closest** to nominal.
- Implemented the methods, analyzed neutrino mass and intensity data, and written the manuscript.

Exact Confidence Intervals for Poisson Signals with Unknown Background

- Designed and implemented iterative confidence-interval refinement using the **h -function**, **Poisson quantile truncation**, and **Berger-Boos correction** to address infinite sample spaces and nuisance-parameter optimization.
- Conducted simulation comparisons of coverage and interval length, and analyzed KARMEN neutrino-detection data.
- Wrote the manuscript and worked on methodological clarification, additional experiments.

PUBLICATIONS

1. Lu, H., & **Wu, Q.** (2026). Power-Enhanced Cauchy Combination Tests for the Two-Sample High-Dimensional Behrens-Fisher Problem. (Under review at *Journal of Multivariate Analysis*)
2. Lu, H., Zhao, S., & **Wu, Q.** (2026). Optimal exact confidence intervals for Poisson signal with unknown background: A quantile-truncated h -function approach with Berger-Boos correction. (Major revision at *Statistical Papers*)
3. Lu, H., & **Wu, Q.** (2026). Prior-free probabilistic interval estimation for constrained parameters. *AIMS Mathematics*, 11(6), 17320–17353. DOI: 10.3934/math.2026709.

PROJECTS

SimForge-GPU: AI-Assisted GPU Migration and Validation for Statistical Simulations	GitHub
• Developed a local CPU-to-GPU migration toolkit for single-file statistical simulations, covering code analysis, conversion planning, code generation, syntax checking, and execution validation. • Used Python AST analysis to identify calls and unsupported behavior; reviewed structured suggestions from external AI coding agents, applied constrained API rewriting, and generated traceable conversion artifacts.	

- Implemented repeated **CPU/GPU** output comparisons and local runtime measurements, with JSON/Markdown validation, benchmark, and explanation reports; included examples and regression tests for **Monte Carlo estimation**.
- Applied GPU computing to the **IM/NIM Monte Carlo studies** reported in *AIMS Mathematics*: with $M = N = 10,000$, $m = 20$, $\lambda = 1$, $\varepsilon = 3$, and nominal 90% coverage, Python/PyTorch batched tensors with vectorized bisection estimated coverage at 0.9083 in **15 s**, versus 0.9106 in **11 h** using R’s serial **uniroot**.

Service Supply-Demand Gap Identification for Older Adults in Guangzhou under High-Temperature Conditions

- Constructed indicators of **heat exposure**, **older-adult vulnerability**, and **service supply**, framing service-gap identification as a **multi-criteria decision-making** problem.
- Combined **entropy weighting**, principal component analysis (**PCA**), **entropy-weighted TOPSIS**, and **Borda ranking** to assess regional supply-demand differences.
- Identified **Yuexiu**, **Haizhu**, and **Liwan** as areas with pronounced supply-demand mismatches and developed **policy recommendations** accordingly.

TECHNICAL SKILLS

Programming & Computing: Python, R; GPU-based Monte Carlo simulation.
Statistical Methods: High-Dimensional hypothesis testing, Interval estimation.
Research Tools: Git/GitHub, LaTeX, Word; literature review and scientific writing.

HONORS & AWARDS

The First Prize Scholarship	Jan. 2027
12th National Undergraduate Statistical Modeling Competition (Second Prize , Guangdong Province)	Jun. 2026
The Third Prize Scholarship	Jan. 2026
15th MathorCup Mathematical Modeling Competition (Third Prize , Guangdong Province)	Apr. 2025
15th National Undergraduate Mathematics Competition (Third Prize , Guangdong Province)	Dec. 2023